

HTML & Forms – Complete Notes

1. World Wide Web (WWW)

The World Wide Web is a system of interlinked documents accessed using the internet. Web pages are created using HTML and displayed using web browsers.

2. What is HTML?

- **HTML** stands for **HyperText Markup Language**
 - Used to create the structure of web pages
 - HTML files are plain text files
 - File extensions: `.html` (preferred) or `.htm`
 - HTML tells the browser *what* to display, not *how* it looks
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3. History of HTML

- Invented by **Tim Berners-Lee**
 - First website published: **August 6, 1991**
 - Created to share scientific documents using hypertext
 - Standardized by **W3C (World Wide Web Consortium)**
 - Major versions:
 - HTML 4 – 1997
 - XHTML – 2000
 - HTML5 – 2014
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4. HTML Tags

- Tags define HTML elements
- Written inside angle brackets: `<tag>`
- Usually come in pairs: `<tag>` `</tag>`
- Tags are **not case-sensitive**
- Must be **properly nested**

Example:

```
<p>This is a paragraph</p>
```

5. Web Terminology

- **Web Design:** Visual planning and layout of a website
 - **Web Development:** Programming and logic behind websites
 - **Front-end:** What users see (HTML, CSS, JS)
 - **Back-end:** Server-side logic and databases
 - **Client:** Browser
 - **Server:** Hosts websites (Apache, NGINX, IIS)
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6. Anatomy of a Website

- **HTML** → Structure
 - **CSS** → Presentation
 - Content + Structure + Style = Website
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7. Structure of an HTML Document

```
<!DOCTYPE html>
<html>
  <head>
    <title>My Page</title>
  </head>
  <body>
    Page content
  </body>
</html>
```

- `<head>` → Metadata, title
 - `<body>` → Visible content
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8. HTML as a Tree Structure

- HTML elements form a **parent-child hierarchy**
 - Helps browsers understand document layout
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9. HTML Elements

Types of Elements

- **Container elements:** `<p>`, `<div>`
- **Standalone elements:** `<br>`, `<img>`

Attributes

- Provide extra information
- Written in opening tag
- Example:

```

```

10. Doctype

- Declares HTML version
- Must be the first line

```
<!DOCTYPE html>
```

11. Whitespace

- HTML ignores extra spaces and newlines
- Use `<p>` for paragraphs
- Use `<pre>` to preserve formatting

12. Nesting Rules

- Elements can contain other elements
- Rule: **First opened = Last closed**

13. Headings

- `<h1>` to `<h6>`
 - Define content hierarchy, not font size
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14. Links (Anchor Tags)

```
<a href="https://www.google.com">Google</a>
```

- Relative links - Absolute links - Page anchors using `#id`

Link Attributes

- `target="_blank"` → Opens in new tab
- `mailto:` → Opens email client

15. Images

```

```

- `src` is mandatory - No closing tag - Supports `.jpg`, `.png`, `.gif`

16. Relative vs Absolute Paths

- **Relative:** Depends on current directory
- **Absolute:** Full URL with domain

17. Lists

Unordered List

```
<ul>
  <li>Item</li>
</ul>
```

Ordered List

```
<ol type="A">
  <li>Item</li>
</ol>
```

18. Comments

```
<!-- This is a comment -->
```

- Not visible in browser - Useful for documentation

19. HTML Entities

- Used for special characters
 - Examples:
 - `<` → <
 - `>` → >
 - `&` → &
 - ` ` → space
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20. Tables

```
<table border="1">  
  <tr><th>Product</th><th>Price</th></tr>  
  <tr><td>Book</td><td>$20</td></tr>  
</table>
```

21. Inline vs Block Elements

- **Block:** `<p>`, `<div>`, `<ul>`
 - **Inline:** `<span>`, `<a>`, `<img>`
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22. div and span

- `<div>` → Block container
 - `<span>` → Inline styling
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23. Forms

- Used to collect user input
- Data sent to server

Form Tag

```
<form action="url" method="post"></form>
```

Methods

- **GET**: Data in URL
 - **POST**: Data in request body
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24. Input Types

- text
 - password
 - radio
 - checkbox
 - submit
 - reset
 - hidden
-

25. Text Fields

```
<input type="text">  
<textarea rows="2" cols="20"></textarea>
```

26. Buttons

- Submit → sends data
 - Reset → clears form
 - Button → JavaScript action
-

27. Checkboxes

```
<input type="checkbox" value="yes">
```

28. Radio Buttons

- Same `name` → only one selectable
-

29. Drop-down List

```
<select>
  <option>Red</option>
</select>
```

30. Hidden Fields

- Not visible to user
 - Sent to server
 - Often set using JavaScript
-

31. HTML5 Semantic Tags

- `<header>`
- `<nav>`
- `<section>`
- `<article>`
- `<main>`
- `<footer>`

Improve readability and SEO

32. Key Takeaways

- HTML defines structure
 - CSS handles appearance
 - Forms enable interaction
 - HTML5 improves semantics
 - View Source is a great learning tool
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End of Notes